

**Cornell Deep**
IDENTITY PROTECTION PLATFORM · V2

☒ Light ☐ Ocean

Sign in

Create account

User name or e-mail

admin / student1 / you@campus.edu

Password

Scenario S1 · Password only (baseline) 

☒ Trust this device after verify

Authenticate

Cornell Deep School · academic project · data is fictional & local only

شاشة الدخول — التيم Light الافتراضي، ومفتاح التبديل إلى 01-login-light.png إلى Ocean

The image shows a login screen for the Cornell Deep Identity Protection Platform (V2) in the Ocean theme. The interface is clean and modern, with a light blue background. The login form is centered and features a dark blue 'Sign in' button and a light blue 'Create account' button. The form includes fields for 'User name or e-mail' (containing 'admin / student1 / you@campus.edu') and 'Password'. Below these fields is a 'Scenario' dropdown menu set to 'S1 - Password only (baseline)' and a checkbox for 'Trust this device after verify' which is checked. A large dark blue 'Authenticate' button is at the bottom of the form. The Cornell Deep logo and name are at the top left, and the version 'V2' is at the top right. The footer text reads 'Cornell Deep School · academic project · data is fictional & local only'.

Cornell Deep
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Light Ocean

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شاشة الدخول بثيم Ocean (أخضر/أزرق) — نفس المحتوى، هوية بصرية مختلفة — login-ocean.png-02

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Scenario S1 · Password only (baseline) ☐ Trust this device after verify

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Cornell Deep School · academic project · data is fictional & local only

login-seed-accounts.png-03 — الحسابات الجاهزة للتجربة مع رمز Demo OTP اللحظي لكل حساب


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☐ Light
 ☒ Ocean

[Sign in](#)
[Create account](#)

Registration uses **real e-mail verification** (Nodemailer / Ethereum https). The verify link stamps this browser as a **Trusted Device** via an encrypted cookie.

Full name

E-mail

National ID

Password (min 8)

Register

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إنشاء حساب جديد ثم التحقق من البريد — بداية رحلة التسجيل — register.png-04

Cornell Deep

ADMINISTRATOR

Portal

Security Center

SOC Console

Forensics

Benchmark

Cornell Deep

SECURITY CENTER

Overview

Audit Log

Sessions

Trusted Devices

Threat Factors

Blacklist

Attack Simulation

Defense OrchestratorADMINISTRATOR

1

SUCCESS

0

FAILED

0

INVALID OTP

0

BLOCKED IPS

0

STEP-UP

0

ALERTS

0

SSE LIVE

Threat overview

updated 09:39:52 PM

LOCKOUT POLICY

2 consecutive failures → immediate block

RISK ENGINE

LOW ≤30 · MED 31–69 · HIGH ≥70

CROSS-DEVICE APPROVAL

WebSocket/SSE push · 45 s TTL · OTP fallback

THREAT FEED

1 recent events

Realtime security feed

LIVE

TIME	OUTCOME	USER	ROLE	RISK	IP · NETWORK	REASONS / MFA
09:39:52 PM	SUCCESS	admin	ADMINISTRATOR	55	127.0.0.1 · Localhost Loopback	NEW_IP, UNRECOGNIZED_DEVICE

Attack simulation timeline

idle

Run an attack from the Attack Simulation tab.

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Light

Ocean

Alex Morgan

Log out

Cornell Deep
SECURITY CENTER

Defense Orchestrator

ADMINISTRATOR

1

SUCCESS

0

FAILED

0

INVALID OTP

0

BLOCKED IPS

0

STEP-UP

0

ALERTS

1

SSE LIVE

[Overview](#)
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Audit log

outcome: all

role: all

user filter

Filter

Clear threat factors

TIME	OUTCOME	ACTOR	RISK	DETAIL
Sep 28, 09:39:52 PM	SUCCESS	admin (ADMINISTRATOR)	55	NEW_IP, UNRECOGNIZED_DEVICE


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Ocean

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Log out


Cornell Deep
SECURITY CENTER

Defense Orchestrator
ADMINISTRATOR

1

0

0

0

0

0

1

SUCCESS

FAILED

INVALID OTP

BLOCKED IPS

STEP-UP

ALERTS

SSE LIVE

Overview
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Active sessions

terminate any session remotely

USER	IP	NETWORK	MFA	STARTED	
admin (ADMINISTRATOR)	127.0.0.1		—	Sep 28, 09:39:52 PM	Terminate



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Defense Orchestrator

ADMINISTRATOR

1

SUCCESS

0

FAILED

0

INVALID OTP

0

BLOCKED IPS

0

STEP-UP

0

ALERTS

1

SSE LIVE

Trusted devices

encrypted trust cookie · 1-year validity

USER	DEVICE	BROWSER	FINGERPRINT	LAST SEEN
user #usr-3001	Edge · Campus PC	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) HeadlessChrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0	fp-769eb667-18c	Sep 28, 09:39:52 PM

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SECURITY CENTER

Defense Orchestrator ADMINISTRATOR

1
SUCCESS

0
FAILED

0
INVALID OTP

0
BLOCKED IPS

0
STEP-UP

0
ALERTS

1
SSE LIVE

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Attack simulation

runs against the REAL auth risk pipeline

Type A · Brute-Force Spray

Launch attack

Live attack telemetry

Run an attack to watch the SOC respond in real time.

Past runs

ID	TITLE	STATUS	EVENTS
----	-------	--------	--------

محاكاة هجوم (SME) — types A/B/C وأثرها الفوري على 09-soc-attack-simulation.png — SOC

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Authentication Mode Comparison

SRS 1.2 — measured from the real audit trail

MODE	ATTEMPTS	AVG LOGIN TIME	SUCCESS	FAILED PASSWORD	2ND-FACTOR FAILURES	BLOCKED	STEP-UPS
Password-only baseline (S1)	0	—	0	0	0	0	0
Password + Mobile OTP MFA (S2)	0	—	0	0	0	0	0
Password + OTP + E-mail step-up + RBAC (S3)	1	25 ms	1	0	0	0	0

Measured from real audit trail (elapsed_ms) — run each scenario ≥3 times for a stable average.

Security / Usability profile (SRS Table 1)

MODE	SECURITY STRENGTH	USABILITY	RISK REDUCED WHEN...
Password-only	Low–Medium	Very easy	adds no protection if the password is stolen
Password + OTP	Medium–High	Easy–moderate	a stolen password alone is insufficient
Password + OTP + E-mail step-up	Higher	Moderate (depends on e-mail access)	an additional barrier if the registered e-mail stays uncompromised

benchmark-scenarios.png-10 — مقارنة السيناريوهات 1/2/3 من سجل حقيقي (المحاولات، النجاح، الزمن)

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Digital Forensic Investigations

timeline reconstruction from the persisted database

CASES
0

AUTH EVENTS
1

ALERTS
0

SUCCESSFUL LOGINS
1

ATTACK RUNS
0

TRUSTED DEVICES
1

Reconstruct timeline

correlated chain across audit · sessions · alerts · approvals · simulations

username

IP address

device fingerprint

outcome: all

Search

Clear

Search any user, IP or device to reconstruct the event chain.

Open a case

wrap a scope into an investigation · auto-captures matching evidence

case title (e.g. Impossible-travel in

CRITICAL

scope: username

scope: IP

scope: device

Create case

Cases

REF	TITLE	SEVERITY	STATUS	EVENTS	CREATED
No cases yet — open one above.					

forensics-timeline.png-11 — إعادة بناء الخط الزمني وفتح قضية مع التقاط الأدلة تلقائيًا

Portal

Security Center

Light

Ocean

Sarah Chen Log out

My Security Center

STUDENT

1

TRUSTED DEVICES

1

ACTIVE SESSIONS

1

ACTIVITY

E-MAIL VERIFIED

Verified

LAST LOGIN RISK POSTURE

recovery enabled · self-service

NATIONAL ID

SCH-2019001345

My trusted devices

DEVICE	BROWSER	FINGERPRINT	LAST SEEN	
Edge · Campus PC	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) HeadlessChrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0	fp-769eb667-18c	Sep 28, 09:40:13 PM	Revoke

My sessions

IP	NETWORK	MFA	STARTED	
127.0.0.1		—	Sep 28, 09:40:13 PM	this session

Recent activity

TIME	OUTCOME	RISK	REASONS
Sep 28, 09:40:13 PM	SUCCESS	55	NEW_IP, UNRECOGNIZED_DEVICE

Account recovery · MFA reset

self-service via registered identity

username or e-mail

Request recovery code

username to validate

recovery code

Validate recovery code

MFA — 12-student-security-center.png مركز الأمان الشخصي للطالب + مسار الاسترداد وإعادة تعيين



Cornell Deep

STUDENT

Portal

Security Center

Light

Ocean

Sarah Chen

Log out

School portal

fictional modules · role: STUDENT · 3 courses · dummy data

Overview

Courses

Grades & Exams

Assignments

Attendance

COURSES

3

GRADES

3

ASSIGNMENTS

4

ATTENDANCE

4

Your academic overview — grades, assignments and attendance for your enrolled courses.

RBAC enforced server-side: every module request is scoped to your STUDENT session.

13-portal-student.png — عرض الطالب: المقررات والدرجات والحضور (RBAC)

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TEACHER

PortalSecurity Center

LightOcean

Dr. Elena RobertsLog out

School portal

fictional modules · role: TEACHER · 2 courses · dummy data

Overview

Courses

Grades & Exams

Assignments

Attendance

Class rosters

COURSES

2

GRADES

3

ASSIGNMENTS

3

ATTENDANCE

6

Your teaching workload — the courses you deliver and their rosters.

RBAC enforced server-side: every module request is scoped to your **TEACHER** session.

**Cornell Deep**
TEACHER

Security Center

Light

Ocean

Dr. Elena Roberts Log out

School portal

fictional modules · role: TEACHER · 2 courses · dummy data

Overview

Courses

Grades & Exams

Assignments

Attendance

Class rosters

COURSES

2

GRADES

3

ASSIGNMENTS

3

ATTENDANCE

6

Your teaching workload — the courses you deliver and their rosters.

RBAC enforced server-side: every module request is scoped to your **TEACHER** session.